Ember: the long life of a $0.1M_\odot$ star

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Ember is a one-dimensional stellar evolution code that calculates a star's structure, composition and energy transport as it ages. We use it to follow a star of $0.1M_\odot$ through 3.795×10^{12} yr from a static model at the beginning of hydrogen burning. Its central hydrogen mass fraction has fallen to $X = 0.007220$, while the surface remains hydrogen-rich at $X = 0.1731$. A core that transports heat by radiation and conduction contains 85.20% of its mass, surrounded by a convective envelope. Hydrogen burning still supplies almost all the luminosity. The aim is to calculate the transition to a helium white dwarf and subsequent cooling through 1000, 500 and 100 K.

The figures show the evolutionary track through the model at the atmosphere table's 3800 K surface temperature limit. We compare this trajectory with the published $0.1M_\odot$ model of Laughlin, Bodenheimer and Adams (1997; LBA97), using their stated values and approximate curves read from their figures. To make a direct numerical comparison, we have generated an f77-source code that seeks to replicate LBA97's exact assumptions and methods.

1 The star and its evolution so far

The initial hydrogen mass fraction is $X_\mathrm{H} = 0.7$, the initial ^{3}He fraction is zero, and the metal mass fraction is $Z = 0.02$. The metals follow the relative elemental abundances of Grevesse and Sauval's GS98 mixture, with the remaining initial mass in ^{4}He. The calculation holds baryonic mass fixed and omits rotation, magnetism, mass loss and external heating. Its age excludes formation and pre-main-sequence contraction.

Model at 3.795Tyr	512 points
Radius ($R_\odot$)	0.1395
Photon luminosity ($L_\odot$)	0.003654
Effective surface temperature (K)	3800
Central temperature (MK)	10.63
Central density ($\mathrm{g\,cm^{-3}}$)	1919
Central hydrogen mass fraction	0.007220
Surface hydrogen mass fraction	0.1731
Central ^{3}He mass fraction	6.412×10^{-7}
Remaining hydrogen mass ($M_\odot$)	0.01157
Convective mass fraction	0.1480

The plotted sequence contains the initial stellar model and 6835 successive models after accepted time steps, each resolved into 512 mass points. Each time step evolves the abundance profile and solves the stellar structure and energy balance, including energy released or absorbed as the structure changes. The calculation uses fixed input tables: 144 gas atmospheres, 72 equation-of-state composition tables, and Los Alamos ATOMIC radiative opacities for the high-temperature interior [1, 2]. The latter include additional hydrogen-poor mixtures through $X = 0$, used above $T = 5.012 \times 10^5$ K; the lower-temperature opacity tables retain their separate composition limits.

1.1 Comparison of the starting model with observed stars

The starting radius, luminosity and temperature can be compared with nearby low-mass stars and an eclipsing companion very close to $0.1 M_\odot$. The following values are reported by Davis et al. [3], Agol et al. [4], and Ribas et al. [5].

Star	Mass ($M_\odot$)	Radius ($R_\odot$)	$T_{\rm eff}$ (K)	Luminosity ($L_\odot$)
Ember, initial	0.1	0.1259	2767	0.0008371
EBLM J2114–39 B	0.0993 ± 0.0033	0.1250 ± 0.0016	—	—
TRAPPIST-1	0.0898 ± 0.0023	0.1192 ± 0.0013	2566 ± 26	0.000553 ± 0.000019
Proxima	0.120 ± 0.003	0.146 ± 0.007	2980 ± 80	0.00151 ± 0.00008

The strongest comparison is EBLM J2114–39 B: its mass agrees with $0.1 M_\odot$, and Ember's radius is only 0.7% larger, within the reported radius uncertainty. The binary's nearly solar metallicity, [Fe/H] $= -0.001 \pm 0.032$, also makes it relevant to our adopted composition. Its mass and radius are inferred from eclipses and orbital velocities together with an estimate of the primary's radius; its temperature and luminosity are not measured in that study.

Ember falls between the lower-mass TRAPPIST-1 and higher-mass Proxima in radius, temperature and luminosity. These comparisons are less direct: TRAPPIST-1's adopted mass uses an empirical mass–luminosity relation, and the quoted Proxima mass comes from stellar models. Age, composition and magnetic activity also differ between individual stars. Within Ember, roughly ten billion years of hydrogen burning changes luminosity by less than 0.4% and temperature by less than 2 K, so ordinary main-sequence age does not materially change this comparison. Formation and early contraction are excluded. Together, the radius agreement at nearly the same mass and the plausible temperature and luminosity support using this initial stellar structure for the calculation. They do not yet establish percent-level accuracy or select uniquely among the atmospheric and interior physics assumptions.

2 Comparison with the published LBA97 model

The main comparison is with Laughlin, Bodenheimer and Adams [6], here abbreviated LBA97. Their section 3.2 and Figure 1 describe the full hydrogen-burning evolution of a $0.1 M_\odot$ star and its entry into white-dwarf cooling. We use numbers stated in the text and approximate values read from the published figure.

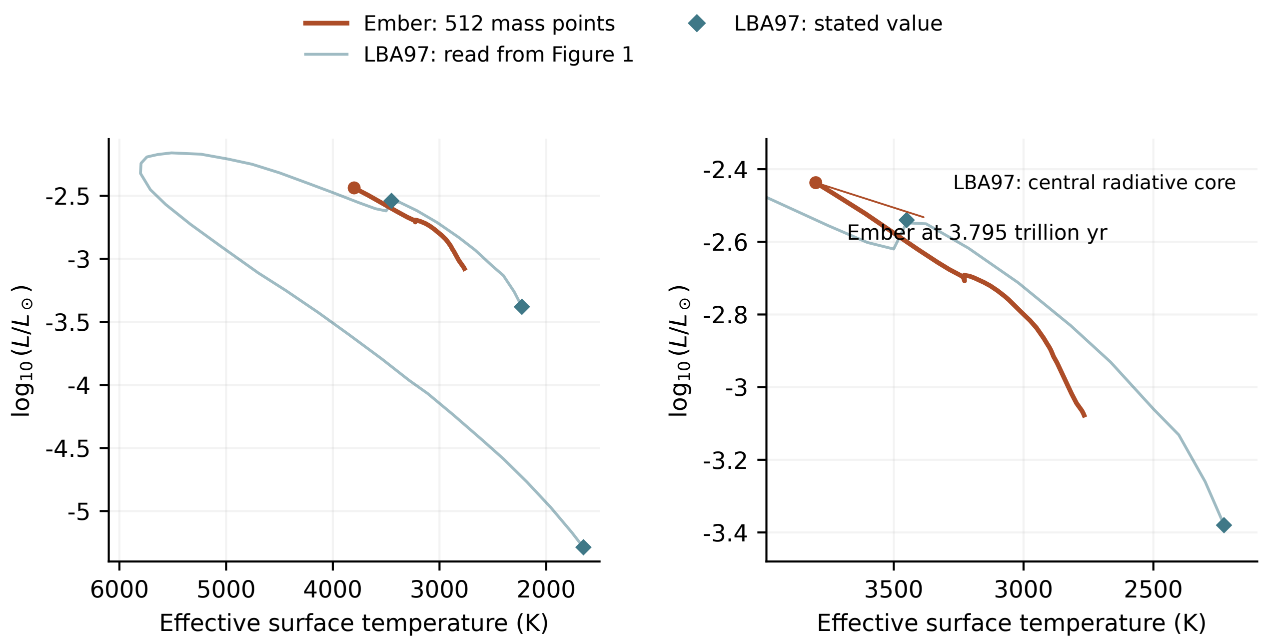

Figure 1: Ember compared with the published LBA97 evolution. The pale curve was read from Figure 1 of LBA97; diamonds mark values stated in their text. The left panel shows the full published path from the start of hydrogen burning through cooling. The right panel enlarges the region reached by Ember. The orange curve contains the complete 512-point Ember calculation through 3.795Tyr. The original pre-main-sequence contraction and the dotted diagonal guide in LBA97 are excluded. The pale line connects 42 selected positions on the printed track; it is an approximate reconstruction, not the original numerical output. Neither temperature nor luminosity has been adjusted to make the curves agree.

Quantity	LBA97, published	Ember
Initial surface temperature (K)	2228	2767
Initial luminosity ($L_\odot$)	$\simeq$ 0.000417	0.0008371
Initial central density ($\mathrm{g\,cm^{-3}}$)	309	382.0
Maximum ^{3}He mass fraction	0.0995	0.1002
Age at maximum ^{3}He (Tyr)	1.38	0.7109
Central radiative core	Forms at 5.742Tyr	Present at 3.795Tyr
Hydrogen mass fraction at core onset	$\simeq$ 0.16	Not yet measured
Highest surface temperature (K)	5807	Not yet reached

The two starting states are defined differently. LBA97 includes contraction and defines main-sequence arrival by nuclear reactions supplying all surface luminosity, after about 2 billion years. Ember starts from a static hydrogen-burning model at age zero. That clock offset is small compared with the later differences, but the initial thermal and atmospheric models also differ. Ember initially radiates about twice the published luminosity and has a hotter surface. These comparisons should not be interpreted as a measurement of the error in one isolated physical ingredient.

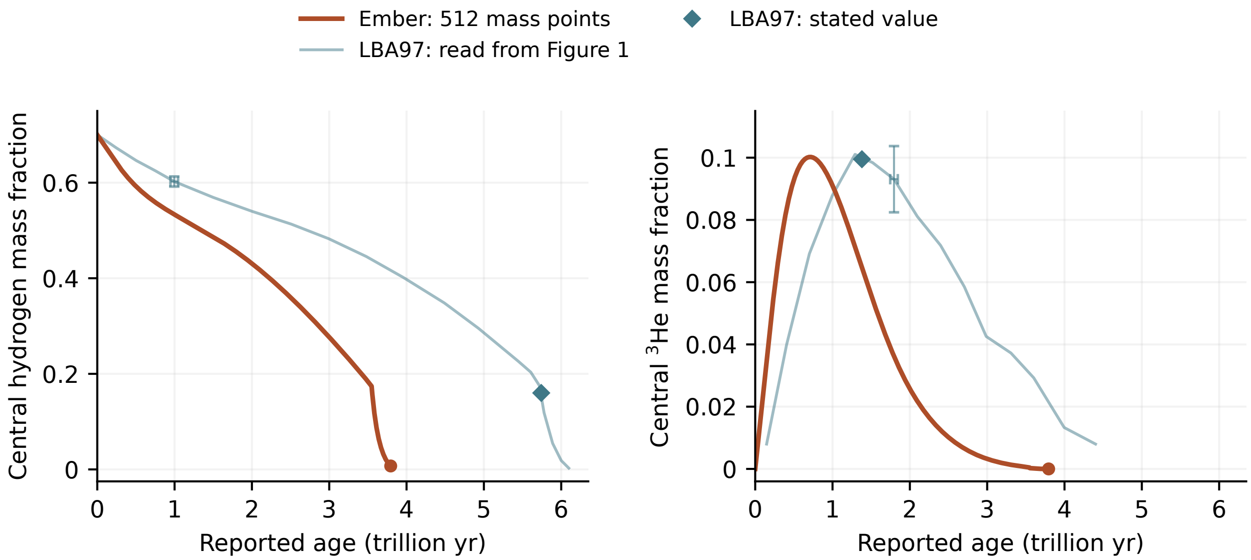

Figure 2: Hydrogen consumption and the buildup and depletion of ^{3}He. The pale curves are read from the inset of LBA97 Figure 1, with 20 hydrogen and 15 helium-3 samples. Diamonds show their stated ^{3}He maximum and the hydrogen abundance at central radiative-core formation. Orange curves are the 512-point Ember track through 3.795Tyr. Ages are shown as reported, without shifting or stretching either calculation. The representative crosses correspond to four image pixels: about 0.05Tyr in age and 0.01 in mass fraction. They illustrate reading precision, not statistical or physical uncertainty. The rapid final hydrogen decline is poorly resolved in the printed inset; the helium-3 trace ends as it becomes indistinguishable from the baseline.

The maximum ^{3}He abundance agrees to within about 1%, but Ember reaches it in about half the time. Ember also consumes hydrogen faster over the interval calculated so far. Reading the published hydrogen curve gives an age of roughly 5.6Tyr at hydrogen fraction 0.2034, reached by Ember near 3.40Tyr. The model at 3.795Tyr has central hydrogen below 0.16, the approximate abundance stated by LBA97 at its central-core transition. The figure does not justify a more precise age comparison. The luminosity–temperature track alone can conceal this difference because it does not show how long the star spends at each position.

LBA97 used a grey atmosphere, approximate opacity changes as hydrogen is consumed, and older interior opacities and nuclear rates. Ember uses frequency-dependent gas atmospheres, composition-specific opacity tables, a different equation of state and updated pp-chain rates. Grain formation is still omitted from Ember's evolving atmosphere. These differences are candidates for the changed surface properties and hydrogen-consumption rate; comparisons that change one ingredient at a time are needed to separate them. Agreement with LBA97 and computing time alone do not establish physical accuracy.

2.1 Core transport by radiation and conduction

The track first departs from full convection near 3.547Tyr, at central hydrogen fraction $X = 0.1751$. By 3.560Tyr, the convective mass fraction has fallen to 0.6330 and central hydrogen to $X = 0.1525$, while surface hydrogen remains at $X = 0.1747$. This separation of central and surface composition shows that convection no longer mixes the whole star.

At 3.795Tyr, the model has a core containing 85.20% of the mass that transports heat by radiation and conduction, surrounded by a convective envelope. Central hydrogen has fallen to $X = 0.007220$, while the envelope remains near $X = 0.1731$. Radiation and conduction can carry the required energy through this core without convection; the composition gradient also suppresses convection. Conduction supplies 40.83% of the local heat flux at the center, with radiation carrying the remaining 59.17%. Conduction is therefore already significant in the central region. Figure 3 shows the declining convective mass and the final abundance profile on this track. The first loss of full convection does not by itself establish a central radiative core. The final profile establishes the core but does not determine its formation age. LBA97 reports central radiative-core formation at 5.742Tyr.

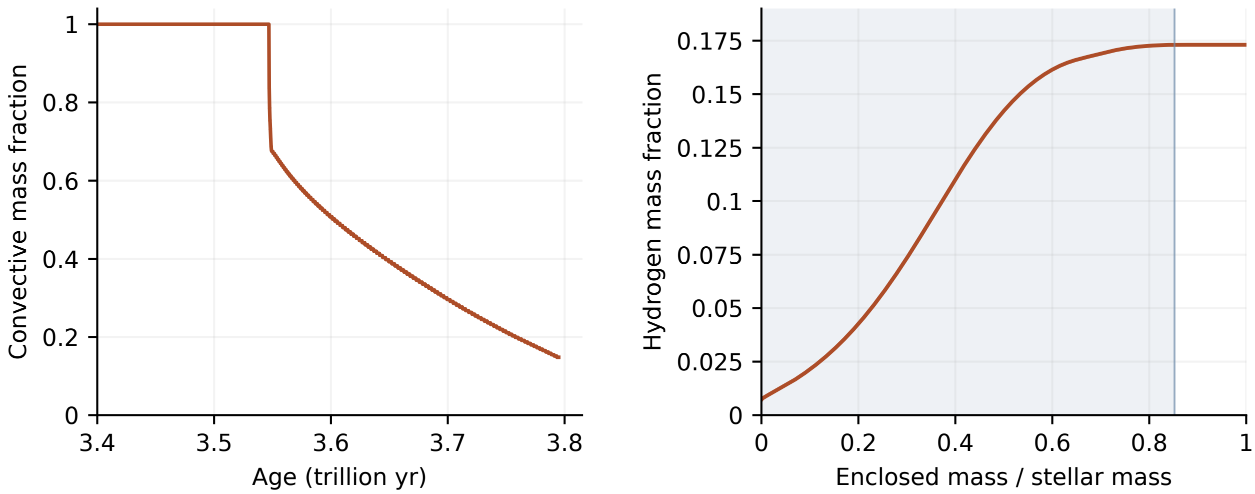

Figure 3: Convection and chemical separation along the Ember track. Left: the fraction of mass in convective regions declines from unity near 3.547Tyr to 0.1480 at 3.795Tyr. Right: the hydrogen profile in the 3.795Tyr model. The shaded core carries heat by radiation and conduction; the outer convective envelope has an almost uniform hydrogen abundance. Both panels use the same calculation and its final saved structure.

"Fully convective" describes where convection occurs, not the fraction of the luminosity it carries. Radiation, conduction and convection share the heat flow continuously. The model's mixing-length calculation makes the convective contribution approach zero at a stability boundary. Chemical mixing uses a stronger approximation: each connected convective region becomes uniform rapidly. Slow composition mixing is included in some marginally stable regions, but their additional heat transport still needs attention.

2.2 The reconstructed F77 program is a separate comparison

The Henyey/F77 program is useful for controlled code comparisons, but it does not reproduce the LBA97 calculation. Its AJR-opacity run first flags the center as nonconvective at 3.224Tyr and $X_{\mathrm{H}} = 0.01756$; its Ferguson-opacity run does so at 2.946Tyr and $X_{\mathrm{H}} = 1.064 \times 10^{-5}$. Both differ substantially from the published 5.742Tyr and $X_{\mathrm{H}} \simeq 0.16$. The AJR run's highest surface temperature, 5936 K, is much closer to LBA97's 5807 K, but agreement in that one quantity does not validate the full history.

The F77 reconstruction substitutes some opacity and nuclear-rate prescriptions. At $X_{\mathrm{H}} = 0.2034$, it is about 76% more luminous than Ember; switching only its low-temperature opacity choice changes its luminosity by 54.5%. These comparisons establish sensitivity to the inputs, rather than identifying the source of the Ember–LBA97 difference. The separate comparison notes and supplementary F77 graphs retain these results. The F77 curves are thin and translucent, with abrupt changes drawn more faintly and all accepted data preserved.

2.3 Photospheric evolution over the opacity table

Figure 4 follows the presentation of LBA97 Figure 6: photospheric density and temperature over an opacity map, with circle diameter proportional to stellar radius. For Ember, we recover the photospheric state at Rosseland optical depth 2/3 from the accepted atmosphere structures. The local temperature there differs from the effective temperature of a frequency-dependent atmosphere. It rises from 3020 to 4111 K over the plotted history, while density falls overall from 1.430×10^{-5} to 6.413×10^{-6} g cm^{-3}. The two opacity maps use the same axes and color scale. The LBA97 map is inferred from the published shading, calibrated against the tabulated opacities of Alexander, Johnson and Rypma [7]. Nineteen cells excluded from the calibration differ by at most 0.12 in $\log_{10} \kappa$; this checks the approximate scale of the diagram, not the accuracy of its grain physics.

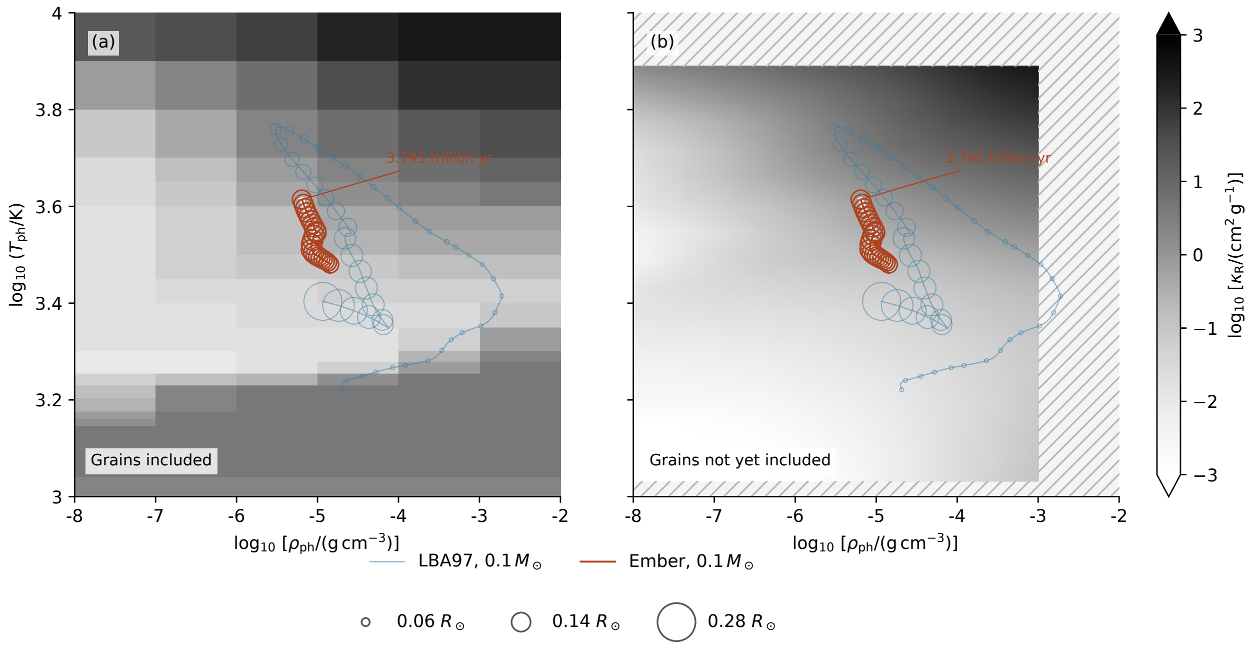

Figure 4: Photospheric evolution of a $0.1 M_\odot$ star over hydrogen-rich opacity maps: (a) the grain-inclusive LBA97 Figure 6 map on an approximate calibrated opacity scale; (b) the Rosseland absorption mean of Ember's SYNSPEC wavelength table at $X = 0.7$, $Z = 0.02$ and zero ^{3}He. Both panels use identical axes and grayscale normalization. The right map excludes scattering and grains; the low-temperature grain-opacity band in the left map is therefore absent. Hatching marks missing wavelength-table coverage. Both panels show the LBA97 track recovered from 897 figure markers and Ember's calculated track through 3.795Tyr. The LBA97 curve includes contraction and cooling; the plotted Ember models are burning hydrogen. Open-circle diameters scale with radius, with 48 LBA97 and 22 Ember samples. LBA97 radii are normalized using its stated main-sequence luminosity and temperature. Ember's local photospheric temperature and density are interpolated at Rosseland optical depth 2/3 using the evolving surface composition. The fixed backgrounds do not represent the changing opacity of either star.

Photospheric temperature and density are interpolated in surface composition, effective temperature and gravity using the atmosphere's molecular equation of state. The corresponding interpolation at the deeper stellar matching boundary reproduces its temperature and pressure to numerical precision at 65 sampled models.

3 Physical ingredients in Ember

Component	Treatment and present limits
Equation of state	FreeEOS 3.0 supplies pressure, internal energy and their responses to temperature, density and composition. Interpolation is based on a shared Helmholtz free energy to keep these responses consistent. The selected 72 composition tables include zero hydrogen and the low-density states needed at the atmosphere boundary. They do not describe the cold, dense remnant.
Opacity	AESOPUS 2.1 provides low-temperature gas opacity, and the LANL TOPS service supplies ATOMIC opacities for hotter and denser conditions [1, 2]. The two are joined only where their available ranges permit it. Hydrogen-poor compositions have been refined and checked against additional calculations. The ATOMIC tables reach zero hydrogen; the cool opacity still requires hydrogen-rich mixtures.
Hydrogen burning	A reduced ppI/ppII network follows ^{3}He out of equilibrium. It uses Solar Fusion II reaction data and finite-degeneracy Salpeter–Van Horn screening, which accounts for the plasma's effect on charged reactants. Atomic mass differences determine heat and nuclear-neutrino energy. Pep, hep, ppIII and CNO reactions are absent; metals do not react.
Heat and composition transport	Mixing-length convection uses the Ledoux criterion, including the stabilizing effect of composition gradients. Slow mixing associated with semiconvection and thermohaline instability is included. Conduction uses approximate mixtures of Ioffe/Cassisi inputs. Microscopic diffusion, settling and a separate semiconvective heat flux are absent.
Atmosphere	TLUSTY208/SYNSPEC54 calculates plane-parallel gas atmospheres in local thermodynamic equilibrium, resolving absorption and scattering over frequency. The interior joins the atmosphere at optical depth 100. The atmospheres include atomic and molecular absorption and collision-induced absorption, but no accepted condensate treatment.
Thermal neutrinos	Plasma-decay losses follow Haft, Raffelt and Weiss [10] and enter the energy equation with analytic derivatives. Other thermal-neutrino processes are not included in this choice.
Numerical evolution	Coupled implicit equations solve structure, burning and mixing. Comparing a full time step with two half steps controls time accuracy. The mass mesh is fixed. Saved models include enough internal information to continue with the same executable, tables and numerical settings.

The interior equation of state, atmosphere chemistry, opacity and conduction come from different physical sources. Their agreement therefore needs checking; accurate derivatives within one component do not prove that all components describe the same thermodynamics. FreeEOS omits potassium from the GS98 mixture, a missing mass fraction of 4.56×10^{-6}. Helium-isotope and atomic partition-function approximations also remain.

4 Why more atmosphere calculations are needed

The atmosphere calculations are reusable. They are performed separately and stored as tables of boundary temperature and pressure. During stellar evolution, Ember interpolates those tables; it does not solve frequency-dependent radiative transfer at every time step. New atmosphere calculations are needed when the star reaches a composition, surface temperature or gravity not adequately covered, or when the physical treatment changes.

The plotted track uses 144 gas atmospheres, spanning the combinations of surface temperature, gravity and composition that it visits through 3800 K. The tabulated source states include hydrogen fractions from 0.1 to 0.7 and ^{3}He fractions from zero to 0.12. These outer ranges do not imply that every combination has been calculated: interpolation requires a complete set of surrounding source models, including those needed for derivatives.

Independent atmosphere calculations test the interpolation. At two 3700 K models with compositions between the tabulated values, matching temperatures differ by −0.4837% and −0.7399%, and gas pressures by +0.5707% and +0.5137%. Changing the lower boundary of the source atmospheres changes matching values by at most 0.007883% in the specified comparisons. All 144 source models have support in the selected 72-table equation of state. Independent comparisons with FreeEOS at 1188 low-density points give a maximum heat-capacity difference of 0.2952%; 792 comparisons at higher densities also pass the stated criteria.

A 156-model atmosphere table extends to 4000 K. At an independently calculated 3900 K atmosphere, interpolated matching temperature differs by −0.4427% and gas pressure by +0.4107%. The largest change in four lower-boundary comparisons is 0.005127%. Chemistry checks find no condensates in the twelve added 3800 K structures or the twelve added 4000 K structures.

The high-temperature ATOMIC opacity tables include additional mixtures with $X = 0$, 0.025, 0.05 and 0.075 at three metal fractions. Twelve independent composition comparisons on hot stellar-profile states differ by at most 0.05257%. This extends the composition coverage needed as central hydrogen is depleted; the lower-temperature tables have their own composition limits.

4.1 Condensates and a hydrogen-free atmosphere

The current evolving atmosphere uses gas-only opacity. Independent chemistry checks found no condensates in the tested 3000–3200 K atmospheres, but found them in the cool upper layers of 2600–2800 K models. Effective temperature alone is therefore insufficient to decide whether grains form. Checks on the added warm hydrogen-poor structures through 3800 K find no condensates; colder structures still require atmosphere calculations with grain formation and opacity included.

Grain formation can affect radiation in two ways: it removes absorbing atoms and molecules from the gas, and any suspended grains absorb and scatter light. A calculation that isolates gas depletion assumes the condensed material settles out. At solar composition and 2800 K, gas depletion changes the interior matching pressure by 0.1246% and temperature by −0.02347%. These are local gas-depletion effects, not measurements of grain opacity or of a stellar lifetime change. Dusty atmosphere models become relevant around 2800 K and below [11]. A separate grain-opacity calculation uses published optical data and explicit grain-size choices. It passes comparisons with an independent scattering program and analytic small-particle limits. Applied to a fixed gas atmosphere, it combines the optical properties with the computed alumina abundance. The maximum condensed mass fraction is 1.094×10^{-4}. For compact grains of radius $0.1\,\mu$m, the integrated grain optical depths near $1\,\mu$m are 0.002610 for absorption and 0.002829 for scattering. These values describe one fixed 2800 K gas profile with suspended grains, not a recomputed atmosphere or a prediction of grain size. Absorption and scattering remain separate. Coupling these opacities to the atmospheric structure requires consistent material phases, wavelength coverage and thermal response, as described in the grain-opacity notes.

Grain formation also changes thermodynamics through redistribution of atoms among molecules, ions and solids. Comparing gas-only and gas-plus-grain chemical equilibria at one test state gives a heat-capacity correction of 1.175×10^5 erg g^{-1} K^{-1}, compared with 2.890×10^5 from counting grains alone. A consistent thermodynamic reference, abundance normalization and treatment across phase boundaries are required before this contribution can enter the evolving model.

The atmosphere wavelength-opacity tables cover material temperatures through 1.315×10^4 K. Nonfinite absorption near He I 4471 Å prevents use of the present source at 1.500×10^4 K. Atmosphere calculations must remain within the valid opacity domain, with matching conditions insensitive to the chosen lower boundary.

A hydrogen-free atmosphere presents a further issue. The source programs' abundance normalization and molecular initialization currently depend on hydrogen. A direct opacity test at 4017 K shows that multiplying all number abundances by the same factor preserves composition and nearly preserves electron density, but can change individual absorption coefficients by a factor of 27.3. This one-temperature test does not measure a stellar-boundary error; it shows that a simple abundance rescaling is not yet a verified solution. The hydrogen-free limit therefore requires a source formulation that does not normalize abundances to hydrogen.

5 Energy and the cold helium remnant

The discrete energy equation used for stellar evolution is

$$L_\gamma = L_{\mathrm{nuc,dep}} + L_{\mathrm{grav}} - L_{\nu,\mathrm{thermal}}, \qquad L_{\mathrm{nuc,rest}} = L_{\mathrm{nuc,dep}} + L_{\nu,\mathrm{nuclear}}.$$

Here L_γ is the photon luminosity, $L_{\mathrm{nuc,dep}}$ is nuclear heat left in the star, and L_{grav} is the contribution from changing internal and gravitational energy. Nuclear and thermal neutrinos are counted separately. At 3.795Tyr in the 512-point model, in solar luminosity units,

$$L_{\mathrm{nuc,dep}} = 0.003650, \quad L_{\mathrm{grav}} = 4.098 \times 10^{-6}, \quad L_{\nu,\mathrm{nuclear}} = 8.838 \times 10^{-5}, \quad L_{\nu,\mathrm{plasma}} = 2.050 \times 10^{-9}.$$

The changing-energy term refers to the last implicit half step; the other reported diagnostics use the saved state. Before rounding, the discrete fractional luminosity residual at this saved endpoint is below 2×10^{-11}. This checks the equations implemented in the code, not an independent exact energy integral or the absence of missing physical processes.

Cold degenerate electrons have a small thermal response superimposed on a large temperature-independent contribution. Paired numerical integration around the Fermi surface retains the small response without subtracting nearly equal large terms. Pure-helium tests at 100–10^4 K and densities 10^3–10^6 g cm^{-3} agree with the expected low-temperature limit to about 10^{-10} fractionally at 100 K. Electron entropy and ionic translational entropy, including mixing, are also implemented and checked against thermodynamic identities. These analytic components have not yet been combined into the production equation of state for a remnant.

Liquid and solid free energies and their derivatives are compared with Ioffe EOSFI22. Across sixteen forced-phase tests, the largest checked density-response discrepancy is 1.05×10^{-6} in ideal-ion pressure units. This tests derivative consistency within each phase; phase selection requires comparing the free energies themselves.

Helium quantum melting, mixture effects, latent heat and the connection to partially ionized outer layers remain unfinished. The source's default classical melting value, $\Gamma = 175$, cannot stand in for a helium phase calculation [12]. Dense, cool atmospheres also require nonideal chemistry and pressure-dependent absorption, including collision-induced absorption; refraction and dense-fluid effects need checking [13].

6 Hydrogen exhaustion and remnant cooling

Extending the track through hydrogen exhaustion requires atmosphere and interior inputs covering the changing composition, temperature and density, including hydrogen-free mixtures and grain formation where relevant.

As composition becomes stratified, conservative microscopic diffusion and settling must include the electric field that maintains charge balance and the associated energy exchange. Moving convection boundaries and thin burning layers need enough mass resolution; an adjustable mesh is required if the fixed mesh does not converge. The importance of omitted nuclear reactions, thermal-neutrino processes and semiconvective heat transport must be assessed in the conditions actually reached.

Hydrogen exhaustion and cooling require distinct measurements. We will record central hydrogen fractions crossing 10^{-3}, 10^{-4} and 10^{-5}, retaining the accepted models on either side. We will also record when nuclear heat falls below 10%, 1% and 0.1% of photon luminosity during cooling, including any later recovery. Remaining hydrogen mass and shell burning must be followed; a depleted center alone does not establish that burning has ended everywhere.

A consistent liquid/solid helium equation of state, conductive transport and a dense atmosphere must allow the star to evolve through 1000, 500 and 100 K on the descending-temperature branch. These endpoints must converge as mass resolution, time accuracy and physical tables are refined. The resulting cooling ages depend on the thermal reservoir, phase changes and atmospheric boundary as well as the numerical accuracy.

The present calculation assumes fixed mass and no external heating. Encounters, irradiation and hypothetical particle-physics effects would alter the very late evolution and require separate physical scenarios [8].

7 Computing time and reproducibility

The plotted 512-point calculation, from the initial model to the atmosphere limit at 3.795Tyr, used 47.27 CPU minutes, accepted 6835 steps and rejected 284 attempts. Elapsed time while the computer was awake was 29.23 minutes; the UTC interval was 61.78 minutes, including suspension. Atmosphere generation is excluded from the stellar CPU time. The calculation reuses exactly matching nuclear responses and shares electron calculations within each response. This is a measured partial trajectory; the cost of hydrogen exhaustion and cooling remains unmeasured.

Repeated nuclear calculations are an identified production cost. Enlarging the cache of exactly matching stellar states reduced CPU time by 42.21% in a short alternating comparison. Sharing the electron calculation within each nuclear response saved a further 14.97% in a separate short comparison with the enlarged cache. All full output files in each comparison agree exactly; 2048 saved late stellar states also give identical nuclear responses and derivatives with the additional electron reuse. Both forms of reuse are included in Ember. A fixed-input timing comparison through 3.757Tyr isolates the larger cache with identical physical inputs and entire output files: CPU time falls from 162.4 to 101.8 minutes, a reduction of 37.31%. The additional electron reuse was not part of that pair. Concurrent background load varied, so this single long pair is not an isolated-machine timing guarantee. The savings during shell burning and remnant cooling remain unmeasured.

The unmodified F77 AJR run used 27.37 CPU seconds through its programmed cooling stop. Ember's finer mass mesh, coupled implicit solves, time-step checks and different physics perform additional work. Their separate contributions to the full speed difference have not been measured. An atmosphere replay spent about 54% of its measured time in chemistry and only 0.0033 seconds in the dense matrix solve; that describes one atmosphere calculation, not the stellar integrator.

The GitHub repository contains code, table-generation instructions, small records of input choices and checks, this paper and the numerical data needed for its figures. The LBA97 comparison records the selected image coordinates, axis calibration and original PDF checksum; its curves can be rebuilt without the original image. The F77 figure data are extracted from accepted models and checked against the recorded output hashes. Full raw outputs, generated physical tables, saved stellar states and executable archives remain local. Instructions for recreating the figures and recovering the recorded stellar calculation accompany the paper files.

Ember passes all 32 test suites with the installed physical tables. The repository supplies figure data, source specifications and validation records; regenerating the bulk physical tables is required for stellar calculations from a fresh checkout.

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